COMPLEX RELATIVITY / RESEARCH ARTICLE

Fundamental Dynamics of Complex Relativity

Closure of the P1–P2 Constraints

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Abstract

The light-cone postulate of Complex Relativity restricts the induced complex metric to $\gamma = (1 + \mathrm{i}\chi)g$, while its momentum postulate identifies a single real free-particle mass shell. These statements become dynamical only after a variational problem has been specified. We construct a real multiplier action implementing the geometric restriction and prove its equivalence, on the regular branch, to variation within the allowed configurations. For the chosen complex curvature action, the reduced geometric theory is exactly Einstein gravity with a canonical scalar. We derive the reduced Hamiltonian and its constraint algebra, supply an explicit auxiliary canonical representation of the geometric restriction, and prove preservation of the compatible complex mass shell for free probes. Positive normal response sectors give an interacting Einstein–scalar system with a common characteristic cone and a local Cauchy development. An exact cosmological solution shows why this constrained theory is not a consistent restriction of the unmodified complex vacuum Einstein equations. A positive deformation of the response metric then shows that P1, P2, and classical closure do not uniquely select the response interactions. The results establish an exact classical realization on the physical real slice and identify the additional selection problem that a fundamental completion must address.

Scope: The actions below are treated as exact classical models. The closure arguments use no expansion in energy or derivatives. Quantum constraint closure and the dynamics away from the physical real slice remain separate problems.

Keywords: Complex Relativity; constrained variation; real mass shell; Hamiltonian constraints; light cone; nonlinear response geometry.

CR manuscripts: github.com/Gauss-K/Complex-Relativity

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Reading guide. Sections 3–5 settle the variational definition. Sections 7–9 establish classical constraint closure and the free-particle result. Sections 10 and 11 give, respectively, the local evolution statement and the limit of what closure can select. The appendix contains component calculations and a reproducibility record.

1 The dynamical question

Suppose the imaginary part of a spacetime metric is proportional to the metric read by real clocks on one initial hypersurface. What prevents the next instant of evolution from generating an independent imaginary tensor component? An algebraic formula on a hypersurface cannot answer that question. The equations of motion and the allowed variations must be known.

Complex Relativity (CR) places real observers on a maximally real four-dimensional slice of a complex spacetime. Its first physical postulate, P1, relates the measured Lorentzian light cone to the complex interval. Its second, P2, relates measured real momenta to a compatible complex quadratic mass shell. Earlier CR manuscripts derive the conformal form of the induced complex metric and construct real particle and field actions [1]. The present article asks which parts of that structure are preserved by a specified interacting classical dynamics.

The word *closure* has several uses here. The geometric restriction must define a legitimate variational problem. Hamiltonian constraints must be preserved under evolution. A complex representation of a real particle constraint must not generate an extra independent mass-shell equation. Finally, the characteristic equations of propagating fields must be compared with the light cone used in P1. These are related tests, but none can replace the others.

Our starting point is a choice already made in the CR construction: the light-cone condition restricts the fundamental configuration space. We implement this choice by multipliers and compare it with restricted variation. This provides an exact classical model. It does not require the restricted solutions to obey the equations of an independently varied complex vacuum metric. Section 5 gives a counterexample to that stronger requirement.

The Hamiltonian calculation uses the ADM formulation of gravity [3, 4]. Its reduced algebra is the familiar hypersurface-deformation algebra, now with the CR scalar and normal response fields included. We distinguish this calculation from a direct Dirac–Bergmann analysis of every unreduced complex-metric momentum. An explicit auxiliary canonical extension represents the same physical reduced phase space, and its second-class constraints can be checked directly. The existence of this representation does not identify its auxiliary momenta with the momenta of the unreduced multiplier Lagrangian.

The final result is a limitation as well as a construction. A whole family of positive response metrics passes the same classical tests. We exhibit members with different intrinsic target curvature. Thus the tests establish consistency of the chosen classical realization; an additional principle is needed to select its particular interactions. Keeping that distinction explicit is necessary if CR is to develop as a candidate fundamental theory.

2 Geometric setting and the two postulates

2.1 Fields on the physical slice

Let M be an oriented, time-oriented four-manifold. The real metric g has signature $(-+++)$, and b is a real symmetric covariant tensor. On the real slice the complex metric is

$$\gamma_{\mu\nu} = g_{\mu\nu} + \mathrm{i}b_{\mu\nu}. \tag{2.1}$$

It is complex bilinear, with no conjugation in its contraction with tangent vectors. We work where g is Lorentzian and γ is invertible. The scalar curvature $R[\gamma]$ below is constructed from the complex Levi–Civita connection of the induced metric on M.

Maximal reality means that, at a point of the slice, a complex tangent has a unique expression $v + \mathrm{i}w$, with $v, w \in T_xM$. It supplies the natural complexification of a real covector. It does not determine the transverse evolution of the ambient complex geometry. All field equations proved here are equations on M.

We use natural units and retain the reduced Planck mass M_{P}. The response parameter is always uppercase C. For the canonical proofs, spatial hypersurfaces are compact without boundary; compactly supported smearings give the same local identities on an open slice. Boundary charges require a separate treatment.

2.2 P1: clocks, inertial comparison, and the shared light cone

P1 assigns clock readings to real timelike observers and requires equivalent local free kinematics in oriented Lorentz inertial frames. Its geometric compatibility clause is

$$g(v,v) = 0 \quad \Longrightarrow \quad \gamma(v,v) = 0, \qquad v \in T_xM \setminus \{0\}. \tag{2.2}$$

Free motion is straight in the flat inertial limit; curved evolution is supplied by dynamics compatible with the real observation structure. Equation (2.2) is an additional compatibility postulate about the complex interval, not a consequence of timing light with clocks coupled only to g.

Lemma 2.1 (Pointwise light-cone rigidity)**.** *For a real symmetric b and a nondegenerate Lorentz metric g in four dimensions, Eq. (2.2) is equivalent to*

$$b_{\mu\nu} = \chi g_{\mu\nu}, \qquad \chi = \frac{1}{4} g^{\mu\nu} b_{\mu\nu}, \qquad \gamma = Fg, \quad F = 1 + \mathrm{i}\chi. \tag{2.3}$$

Proof. Choose a g-orthonormal frame and write a future null vector as $(1, \boldsymbol{n})$, with $|\boldsymbol{n}| = 1$. Evaluating b on $(1, \pm e_i)$ gives $b_{0i} = 0$ and $b_{ii} = -b_{00}$. Evaluation on $(1, (e_i + e_j)/\sqrt{2})$, for each $i < j$, then gives $b_{ij} = 0$. Hence $b = \chi g$, with $\chi = -b_{00}$. The trace gives the coordinate-independent formula. Conversely a multiple of g vanishes on the real light cone. Since $F \neq 0$ for real χ, the real null sets of γ and g coincide. □

The trace-free tensor

$$E_{\mu\nu} = b_{\mu\nu} - \frac{1}{4}(g^{\rho\sigma} b_{\rho\sigma}) g_{\mu\nu} \tag{2.4}$$

has nine independent components. P1 sets these components to zero. This is a count of algebraic conditions, not a count of propagating modes or gauge generators.

2.3 P2: real momenta and a compatible complex shell

P2 takes the free-particle momentum to be a real covector p, the invariant mass to be real with $m \geq 0$, and the physical shell to be

$$\mathcal{Q}_{\mathrm{R}} = g^{\mu\nu} p_\mu p_\nu + m^2 = 0 \tag{2.5}$$

on the energy branch selected by time orientation. Momentum extends naturally by

$$p^{\mathbb{C}}(v + \mathrm{i}w) = p(v) + \mathrm{i}p(w). \tag{2.6}$$

The full complex quadratic shell must represent the same free states, reduce to the reference shell at $\chi = 0$, and introduce no second independent constraint on real momentum. Free evolution uses the real cotangent symplectic structure.

For a momentum-independent mass coefficient, these requirements give

$$\mathcal{Q}_{\mathbb{C}} = \gamma^{-1}(p^{\mathbb{C}}, p^{\mathbb{C}}) + F^{-1} m^2 = F^{-1} \mathcal{Q}_{\mathrm{R}}. \tag{2.7}$$

For $m > 0$, substitution of the complete real mass shell fixes the coefficient to F^{-1}. At $m = 0$, the unused mass coefficient is not determined. The particle constraint is regular except at the massless vertex $p = 0$, which is excluded in the statements below.

2.4 What is assumed about dynamics

P1 and P2 specify compatibility requirements. They do not select every generally covariant action. We choose the real part of the complex Einstein–Hilbert functional for the geometric core, impose P1 as a configuration constraint, and later couple a specified positive response metric. These choices are displayed in the action. No uniqueness of that action is assumed.

3 An exact constrained variational principle

3.1 The multiplier construction

In a neighborhood of the allowed configurations, choose the branch of $\sqrt{-\det\gamma}$ continuous from $\gamma = g$, and define

$$S_0[g,b] = \frac{M_{\mathrm{P}}^2}{2}\,\mathrm{Re}\int_M \mathrm{d}^4x\,\sqrt{-\det(g+\mathrm{i}b)}\,R[g+\mathrm{i}b], \tag{3.1}$$

$$S_{\mathrm{c}}[g,b,\chi,\lambda] = S_0[g,b] + \int_M \mathrm{d}^4x\sqrt{-g}\,\lambda^{\mu\nu}(b_{\mu\nu}-\chi g_{\mu\nu}). \tag{3.2}$$

Here $\lambda^{\mu\nu}$ is real and symmetric, and χ is initially independent. Variations have compact support; equivalently, suitable boundary data and boundary terms are supplied. This convention isolates the bulk argument from the boundary value problem.

Put $B_{\mu\nu} = b_{\mu\nu} - \chi g_{\mu\nu}$. The ten equations $B = 0$ are regular because their derivative with respect to the ten components of b, at fixed g, χ, is the identity. Adding χ as an independent variable and then imposing ten equations leaves eleven configuration variables: ten components of g and one scalar. Relative to the original twenty variables (g, b), the net reduction is nine.

3.2 Equivalence with restricted variation

Define real Euler derivatives by

$$\delta S_0 = \int_M \mathrm{d}^4x\sqrt{-g}\left(\mathcal{E}_g^{\mu\nu}\delta g_{\mu\nu} + \mathcal{E}_b^{\mu\nu}\delta b_{\mu\nu}\right) + \text{boundary terms}. \tag{3.3}$$

This definition uses covariant metric variations. The Euler derivatives include all variations of the volume factor in S_0.

Proposition 3.1 (Exact multiplier equivalence)**.** *On the regular branch, solutions of S_{c} correspond one-to-one to solutions of*

$$S_{\mathrm{red}}[g,\chi] = S_0[g,\chi g], \tag{3.4}$$

with $b = \chi g$ and $\lambda^{\mu\nu} = -\mathcal{E}_b^{\mu\nu}|_{b=\chi g}$. The correspondence is a statement about the constrained theory, not about the unconstrained equations $\mathcal{E}_g = \mathcal{E}_b = 0$.

Proof. The λ-variation gives $B = 0$. On that surface, the other equations are

$$\mathcal{E}_b^{\mu\nu} + \lambda^{\mu\nu} = 0, \qquad \mathcal{E}_g^{\mu\nu} - \chi\lambda^{\mu\nu} = 0, \qquad g_{\mu\nu}\lambda^{\mu\nu} = 0. \tag{3.5}$$

The variation of $\sqrt{-g}$ in the multiplier term is proportional to $\lambda^{\rho\sigma}B_{\rho\sigma}$ and vanishes on $B = 0$. Eliminating λ gives

$$\mathcal{E}_g^{\mu\nu} + \chi\mathcal{E}_b^{\mu\nu} = 0, \qquad g_{\mu\nu}\mathcal{E}_b^{\mu\nu} = 0. \tag{3.6}$$

Restricted variations satisfy $\delta b = \chi\delta g + g\delta\chi$. Substitution in Eq. (3.3) produces exactly Eq. (3.6). Conversely, any solution of the restricted equations, with the stated b and λ, solves all the multiplier equations. The multiplier is fixed by the other fields and carries no independent solution data. □

The proof is an explicit instance of the distinction between substitution in an action and substitution in its Euler equations [2]. The multiplier supplies the reaction needed in the excluded directions. Requiring that reaction to vanish would define an additional theory.

The correspondence is local on the stated regular branch and uses the same bulk boundary convention on both sides. It concerns the constrained solution space; it does not assert that arbitrary unconstrained initial data evolve toward $B = 0$, or that the unconstrained complex theory preserves that surface.

An interaction $S_{\rm m}[g, \chi, q]$, with no b-dependence, may be added on both sides. The χ-equation then includes its source, and the trace of λ need not vanish. The same chain-rule proof establishes equivalence. We will use this extension for the response fields.

3.3 Physical symplectic structure

The multiplier term contains no derivatives and contributes no boundary symplectic potential. On tangent variations with $\delta B = 0$, the potential from S_0 pulls back to the potential from $S_{\rm red}$. Removing a divergence in the Lagrangian changes the potential by an exact variation, with no change in the integrated symplectic form on a closed spatial slice. Consequently the two descriptions have the same physical presymplectic form on the corresponding solution spaces, modulo diffeomorphisms.

This observation justifies a reduced canonical description. It does not assert that every momentum in the unreduced Lagrangian vanishes when $B = 0$. The distinction matters when constructing auxiliary canonical variables in Section 8.

4 The geometric core and its field equations

For $F = 1 + \mathrm{i}\chi$, the chosen branch satisfies $\sqrt{-\det(Fg)} = F^2\sqrt{-g}$. The four-dimensional conformal identity gives

$$\sqrt{-\det(Fg)}R[Fg] = \sqrt{-g}\left(FR[g] - 3\Box_g F + \frac{3}{2F}(\nabla F)^2\right). \tag{4.1}$$

The contractions on the right use g. Since

$$\mathrm{Re}\,\frac{(\nabla F)^2}{F} = -\frac{(\nabla\chi)^2}{1+\chi^2}, \tag{4.2}$$

the reduced action is

$$S_{\rm red} = \int_M \mathrm{d}^4x\sqrt{-g}\left[\frac{M_{\rm P}^2}{2}R - \frac{3M_{\rm P}^2}{4(1+\chi^2)}(\nabla\chi)^2\right]. \tag{4.3}$$

The real part of $\Box F$ vanishes here; the divergence in Eq. (4.1) is also compatible with the boundary convention already adopted.

Define

$$f^2 = \frac{3}{2}M_{\rm P}^2, \qquad \phi = f\,\mathrm{arsinh}\,\chi, \qquad \chi = \sinh(\phi/f). \tag{4.4}$$

This is a smooth, invertible change of real variables for all finite χ. It yields the exact action

$$S_{\rm red} = \int_M \mathrm{d}^4x\sqrt{-g}\left[\frac{M_{\rm P}^2}{2}R - \frac{1}{2}(\nabla\phi)^2\right]. \tag{4.5}$$

The equations are

$$M_{\rm P}^2 G_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - \frac{1}{2}g_{\mu\nu}(\nabla\phi)^2, \qquad \Box_g\phi = 0. \tag{4.6}$$

Equivalently,

$$R_{\mu\nu}[g] = \frac{3}{2(1+\chi^2)}\partial_\mu\chi\partial_\nu\chi, \qquad \Box_g\chi - \frac{\chi}{1+\chi^2}(\nabla\chi)^2 = 0. \tag{4.7}$$

The scalar has the positive kinetic sign. The count of physical modes still requires the gravitational constraints; it is performed below.

The equality with Einstein gravity and a real scalar is a mathematical equivalence of this chosen core. A change of variables cannot by itself supply new observable predictions. Any extra physical restrictions must appear in the permitted interactions, state space, or additional geometric structure.

5 Why unconstrained complex vacuum evolution is different

Independently varying the real and imaginary components of a nondegenerate complex metric in $\mathrm{Re}\, S_{\mathrm{EH}}[\gamma]$ imposes the full complex vacuum equation. In four dimensions this is equivalent to $R_{\mu\nu}[\gamma] = 0$. It is stronger than the projected equations in Proposition 3.1.

5.1 A linear obstruction

Let $g = \eta + O(\epsilon^2)$ and $\chi = O(\epsilon)$. The constrained theory permits $\Box_\eta \chi = 0$ at first order, while

$$\mathrm{Im}\, R_{\mu\nu}[\gamma] = -\partial_\mu \partial_\nu \chi - \frac{1}{2}\eta_{\mu\nu}\Box_\eta \chi + O(\epsilon^2). \tag{5.1}$$

A nonconstant plane wave $\chi = \epsilon \cos(k \cdot x)$, $k^2 = 0$, generally has nonzero Hessian. It is a permitted linearized scalar mode of the constrained theory and fails the unmodified complex vacuum equation.

5.2 An exact counterexample

Take $t > 0$, a constant $t_0 > 0$, and a flat spatial metric. The fields

$$g = -\mathrm{d}t^2 + a(t)^2 \mathrm{d}\boldsymbol{x}^2, \qquad a(t) = (t/t_0)^{1/3}, \tag{5.2}$$

$$\phi(t) = \sqrt{\frac{2}{3}} M_{\mathrm{P}} \log(t/t_0), \qquad \chi(t) = \sinh\left(\frac{2}{3}\log(t/t_0)\right) \tag{5.3}$$

solve Eq. (4.6). The solution may be considered locally, or with a flat spatial torus. It has $H = 1/(3t)$, $R_{tt}[g] = 2/(3t^2)$, and $R_{ij}[g] = 0$.

For a general F, the complex Ricci tensor is

$$R_{\mu\nu}[Fg] = R_{\mu\nu}[g] - F^{-1}\nabla_\mu \nabla_\nu F - \frac{1}{2}F^{-1} g_{\mu\nu}\Box F + \frac{3}{2}F^{-2}\partial_\mu F \partial_\nu F. \tag{5.4}$$

At $t = t_0$, where $a = 1$ and $\chi = 0$, direct substitution gives

$$R_{tt}[\gamma] = \frac{2\mathrm{i}}{3t_0^2}, \qquad R_{ij}[\gamma] = \frac{2\mathrm{i}}{9t_0^2}\delta_{ij}. \tag{5.5}$$

Thus the constrained solution is not a solution of the unmodified complex vacuum theory, even at an event where $\gamma = g$. Derivatives of the complex factor remain relevant. Appendix E records two independent evaluations of these components.

The reaction multiplier can be displayed at the same event. For the covariant metric variations in Eq. (3.3),

$$\mathcal{E}_b^{\mu\nu} = \frac{M_{\mathrm{P}}^2}{2}\,\mathrm{Im}\left[\frac{\sqrt{-\det\gamma}}{\sqrt{-g}}\,\mathcal{G}^{\mu\nu}[\gamma]\right], \tag{5.6}$$

where $\mathcal{G}^{\mu\nu}[\gamma]$ is the complex Einstein tensor with both indices raised by γ. At $t = t_0$, its complex scalar curvature is zero. Hence

$$\lambda^{tt} = -\frac{M_{\mathrm{P}}^2}{3t_0^2}, \qquad \lambda^{ij} = -\frac{M_{\mathrm{P}}^2}{9t_0^2}\delta^{ij}, \qquad \lambda^{ti} = 0. \tag{5.7}$$

This tensor is nonzero and g-trace-free, as required by Eq. (3.5). The example exhibits the reaction that distinguishes the constrained model from complex vacuum gravity.

Corollary 5.1. *The substitution $\gamma = (1 + \mathrm{i}\chi)g$, with the dynamics of Eq. (4.5), is not a consistent truncation of the unmodified complex vacuum Einstein equations in the sense that every reduced solution lifts to a parent solution.*

This result concerns that specific unconstrained parent. The multiplier theory is different and already supplies the required reaction equations. Other proposed parents must be tested using their own field equations.

6 Normal response fields and the coupled action

6.1 A specified response geometry

Introduce r normal response sectors, with constants $C_a > 1$. The scalar fields are

$$\Phi^A = (\phi, U_1, V_1, \ldots, U_r, V_r), \qquad n = 1 + 2r, \qquad x = \phi/f. \tag{6.1}$$

Use the response matrix of the preceding CR construction,

$$\mathsf{K}_C(\phi) = \begin{pmatrix} \dfrac{C\cosh x + 1}{C+1} & -\dfrac{\sinh x}{\sqrt{C^2-1}} \\ -\dfrac{\sinh x}{\sqrt{C^2-1}} & \dfrac{C\cosh x - 1}{C-1} \end{pmatrix}, \tag{6.2}$$

and the target metric

$$\mathrm{d}s^2_{\mathcal{T}} = \mathsf{G}_{AB}\mathrm{d}\Phi^A\mathrm{d}\Phi^B = \mathrm{d}\phi^2 + \sum_{a=1}^{r} \mathrm{d}q_a^T \mathsf{K}_{C_a}(\phi)\mathrm{d}q_a, \qquad q_a = (U_a, V_a)^T. \tag{6.3}$$

At the reference vacuum, $\mathsf{K}_C(0) = I_2$, and

$$\det \mathsf{K}_C(\phi) = \cosh^2(\phi/f) > 0. \tag{6.4}$$

The first diagonal entry is positive for $C > 1$; hence each block is positive definite. To identify the unnormalized form explicitly, let

$$\mathsf{D}_C = \mathrm{diag}(\sqrt{C+1}, \sqrt{C-1}), \qquad \mathsf{D}_C\mathsf{K}_C\mathsf{D}_C = \begin{pmatrix} C\cosh x + 1 & -\sinh x \\ -\sinh x & C\cosh x - 1 \end{pmatrix}. \tag{6.5}$$

The matrix on the right has eigenvalues $(C \pm 1)\cosh x$. This is a congruence, not a similarity transformation; those are not generally the eigenvalues of the normalized K_C. Positivity holds for finite ϕ and every fixed $C > 1$, with no small-field approximation.

The response matrix is additional dynamical data. The postulates explain how to test its compatibility with real observations; they have not uniquely selected Eq. (6.2). Section 11 makes this distinction quantitative.

6.2 Exact nonlinear equations

After the geometric multiplier reduction, the coupled action is

$$S = \int_M \mathrm{d}^4x\sqrt{-g}\left[\frac{M_{\mathrm{P}}^2}{2}R - \frac{1}{2}\mathsf{G}_{AB}(\Phi)g^{\mu\nu}\partial_\mu\Phi^A\partial_\nu\Phi^B - V(\Phi)\right]. \tag{6.6}$$

The baseline model takes $V = 0$. A smooth potential is retained in the formulas to identify which closure statements are independent of that choice. If the interaction is added to Eq. (3.2), the ϕ kinetic term is supplied by the geometric core and is not added a second time.

Variation gives

$$M_{\rm P}^2 G_{\mu\nu} = T_{\mu\nu}, \tag{6.7}$$

$$T_{\mu\nu} = \mathsf{G}_{AB}\partial_\mu\Phi^A\partial_\nu\Phi^B - g_{\mu\nu}\left[\frac{1}{2}\mathsf{G}_{AB}(\nabla\Phi^A\cdot\nabla\Phi^B) + V\right], \tag{6.8}$$

$$\mathcal{E}^A = \Box_g\Phi^A + \Gamma^A{}_{BC}(\mathsf{G})\nabla\Phi^B\cdot\nabla\Phi^C - \mathsf{G}^{AB}V_{,B} = 0. \tag{6.9}$$

The target connection is evaluated from G. These equations include the source of ϕ from the response sectors and the backreaction of all scalars on g.

Diffeomorphism invariance gives the useful identity

$$\nabla^\mu T_{\mu\nu} = \mathsf{G}_{AB}\mathcal{E}^A\partial_\nu\Phi^B. \tag{6.10}$$

It can also be verified directly using metric compatibility of the target connection. Total stress is conserved on the scalar equations. Individual response blocks exchange energy with ϕ and are not separately conserved in a general background.

7 The reduced Hamiltonian and its closed algebra

7.1 ADM variables and momenta

Write

$$\mathrm{d}s_g^2 = -N^2\mathrm{d}t^2 + h_{ij}(\mathrm{d}x^i + N^i\mathrm{d}t)(\mathrm{d}x^j + N^j\mathrm{d}t), \qquad K_{ij} = \frac{\dot h_{ij} - D_iN_j - D_jN_i}{2N}. \tag{7.1}$$

The lapse $N > 0$ and shift N^i are multipliers. The connection D_i belongs to h. With the usual boundary term understood, the canonical momenta are

$$\pi^{ij} = \frac{M_{\rm P}^2\sqrt{h}}{2}(K^{ij} - Kh^{ij}), \qquad P_A = \frac{\sqrt{h}}{N}\mathsf{G}_{AB}(\dot\Phi^B - N^i\partial_i\Phi^B). \tag{7.2}$$

In the regular normal sector, both relations can be inverted for the physical velocities. The canonical action is

$$S = \int \mathrm{d}t\mathrm{d}^3x\,\left(\pi^{ij}\dot h_{ij} + P_A\dot\Phi^A - N\mathcal{H} - N^i\mathcal{D}_i\right), \tag{7.3}$$

where

$$\mathcal{H} = \frac{2}{M_{\rm P}^2\sqrt{h}}\left(\pi^{ij}\pi_{ij} - \frac{1}{2}\pi^2\right) - \frac{M_{\rm P}^2\sqrt{h}}{2}\,{}^{(3)}\!R + \frac{1}{2\sqrt{h}}\mathsf{G}^{AB}P_AP_B + \frac{\sqrt{h}}{2}\mathsf{G}_{AB}h^{ij}\partial_i\Phi^A\partial_j\Phi^B + \sqrt{h}\,V, \tag{7.4}$$

$$\mathcal{D}_i = -2h_{ij}D_k\pi^{jk} + P_A\partial_i\Phi^A, \qquad \pi = h_{ij}\pi^{ij}. \tag{7.5}$$

Equations (7.4)–(7.5) retain the full target metric. No truncation of its ϕ-dependence has been made.

The nonzero fundamental brackets are

$$\{h_{ij}(\boldsymbol{x}), \pi^{kl}(\boldsymbol{y})\} = \delta^k_{(i}\delta^l_{j)}\delta^3(\boldsymbol{x}-\boldsymbol{y}), \qquad \{\Phi^A(\boldsymbol{x}), P_B(\boldsymbol{y})\} = \delta^A_B\delta^3(\boldsymbol{x}-\boldsymbol{y}). \tag{7.6}$$

Define smeared constraints

$$H[N] = \int_\Sigma \mathrm{d}^3x\,N\mathcal{H}, \qquad D[\boldsymbol{\xi}] = \int_\Sigma \mathrm{d}^3x\,\xi^i\mathcal{D}_i. \tag{7.7}$$

7.2 A direct calculation of the matter bracket

Split $H = H_g + H_\Phi$ and $D = D_g + D_\Phi$. Put

$$v^A = \frac{\mathsf{G}^{AB}P_B}{\sqrt{h}}, \qquad J^i_A = \sqrt{h}\,\mathsf{G}_{AB}h^{ij}\partial_j\Phi^B. \tag{7.8}$$

The functional derivatives have the form

$$\frac{\delta H_\Phi[N]}{\delta P_A} = Nv^A, \qquad \frac{\delta H_\Phi[N]}{\delta \Phi^A} = N\mathcal{A}_A - \partial_i(NJ^i_A), \tag{7.9}$$

where $\mathcal{A}_A$ contains the local target-metric and potential derivatives. Terms proportional to NM cancel under antisymmetrization. The remaining terms give

$$\begin{aligned}\{H_\Phi[N], H_\Phi[M]\} &= \int_\Sigma \mathrm{d}^3x\,(N\partial_i M - M\partial_i N)J^i_A v^A \\ &= D_\Phi[\boldsymbol{\beta}], \qquad \beta^i = h^{ij}(N\partial_j M - M\partial_j N).\end{aligned} \tag{7.10}$$

The identity $J^i_A v^A = h^{ij}P_A\partial_j\Phi^A$ is the only target-metric cancellation needed. It holds for any invertible G.

The antisymmetrized gravity–matter cross bracket vanishes. In this action, H_Φ contains no π^{ij} or spatial derivatives of h_{ij}, while $\delta H_g[N]/\delta\pi^{ij}$ is local and proportional to N. The cross terms are therefore symmetric in NM. Combining Eq. (7.10) with the gravitational bracket gives the complete result.

7.3 The gravitational sign and normalization

The only part of $\delta H_g[N]/\delta h_{ij}$ containing derivatives of the lapse is

$$\left.\frac{\delta H_g[N]}{\delta h_{ij}}\right|_{\partial N} = -\frac{M_{\mathrm{P}}^2\sqrt{h}}{2}\left(D^iD^jN - h^{ij}D^2N\right), \qquad D^2 = D_kD^k. \tag{7.11}$$

Meanwhile

$$\frac{\delta H_g[N]}{\delta \pi^{ij}} = \frac{4N}{M_{\mathrm{P}}^2\sqrt{h}}\left(\pi_{ij} - \frac{1}{2}\pi h_{ij}\right). \tag{7.12}$$

Terms containing no lapse derivatives cancel upon antisymmetrization. Contracting the two displayed expressions cancels their trace terms and gives

$$\begin{aligned}\{H_g[N], H_g[M]\} &= 2\int_\Sigma \mathrm{d}^3x\,\pi^{ij}\,(ND_iD_jM - MD_iD_jN) \\ &= -2\int_\Sigma \mathrm{d}^3x\,(D_i\pi^{ij})\,(N\partial_jM - M\partial_jN) = D_g[\boldsymbol{\beta}].\end{aligned} \tag{7.13}$$

Here π^{ij} is a tensor density of weight one. The integration by parts uses the boundary convention already stated; the term involving the antisymmetric product of first lapse derivatives vanishes against symmetric π^{ij}. This calculation fixes the sign and factors for the convention of Eq. (7.1).

Theorem 7.1 (Reduced classical constraint closure)**.** *For Eq. (6.6), with smooth invertible* G*, smooth* V*, and the boundary convention stated above, the canonical constraints obey*

$$\{D[\boldsymbol{\xi}], D[\boldsymbol{\eta}]\} = D[[\boldsymbol{\xi}, \boldsymbol{\eta}]], \tag{7.14}$$

$$\{D[\boldsymbol{\xi}], H[N]\} = H[\mathcal{L}_{\boldsymbol{\xi}}N], \tag{7.15}$$

$$\{H[N], H[M]\} = D[\boldsymbol{\beta}], \quad \beta^i = h^{ij}(N\partial_jM - M\partial_jN). \tag{7.16}$$

Consequently $\mathcal{H} = 0$ *and* $\mathcal{D}_i = 0$ *are preserved by the total Hamiltonian, for arbitrary lapse and shift within the chosen foliation.*

Proof. The momentum constraint generates spatial Lie derivatives on the canonical tensors and scalars. This proves Eqs. (7.14) and (7.15), since $\mathcal{H}$ is a scalar density of weight one. Equation (7.16) follows from the preceding matter calculation, Eq. (7.13), and cancellation of the cross terms. Bracketing a constraint with $H[N] + D[\boldsymbol{N}]$ therefore yields a combination of constraints. If lapse and shift momenta are included, their vanishing gives four primary constraints whose preservation yields $\mathcal{H}$ and $\mathcal{D}_i$; no new matter constraint arises on the nondegenerate branch. □

This theorem uses the standard gravitational constraint structure [3, 4]; the explicit matter calculation identifies its applicability to the CR response metric. Positive definiteness supplies positive scalar kinetic energy and is assumed in the Cauchy statement below; algebraic closure itself only uses invertibility. The occurrence of h^{ij} in $\boldsymbol{\beta}$ means that the algebra has field-dependent structure functions. Classical first-class closure is not a constant-coefficient Lie algebra or a proof of quantum anomaly cancellation.

7.4 Physical degrees of freedom

There are $6 + n$ canonical configuration variables (h_{ij}, Φ^A). The four first-class secondary constraints remove eight phase-space dimensions. With lapse and shift treated as multipliers, the local count is

$$N_{\text{phys}} = \frac{2(6+n) - 2 \times 4}{2} = 2 + n. \tag{7.17}$$

The equivalent count including lapse, shift, and their momenta gives the same result.

Classical model	Scalar modes	Total physical modes
Geometric core	1	3
Core and one normal response sector	3	5
Core and two normal response sectors	5	7

The two additional modes are the tensor polarizations of gravity. A varying target metric couples the scalars but does not create a gauge constraint when its kinetic matrix is invertible. Potential instabilities must be checked separately from this degree count.

8 A canonical representation of the geometric restriction

P1's nine independent algebraic restrictions should not be identified with nine first-class generators. An explicit canonical representation makes the distinction clear.

Start with the reduced phase space of Section 7. In an adapted local chart, append ten auxiliary pairs (B^a, Π_a), $a = 1, \ldots, 10$, with

$$\{B^a(\boldsymbol{x}), \Pi_b(\boldsymbol{y})\} = \delta^a_b \delta^3(\boldsymbol{x} - \boldsymbol{y}). \tag{8.1}$$

The labels a enumerate the components of $B = b - \chi g$, using χ as the extra scalar coordinate. Impose

$$\Theta_I = (B^a, \Pi_a) = 0. \tag{8.2}$$

Their bracket matrix is

$$\Delta_{IJ}(\boldsymbol{x}, \boldsymbol{y}) = \begin{pmatrix} 0 & I_{10} \\ -I_{10} & 0 \end{pmatrix} \delta^3(\boldsymbol{x} - \boldsymbol{y}), \tag{8.3}$$

which is invertible. They are second class.

Extend H and D independently of these auxiliary variables and use

$$H_{\mathrm{T}}^{\mathrm{ext}} = H[N] + D[\boldsymbol{N}] + \int \mathrm{d}^3x\,(u_a B^a + v^a \Pi_a). \tag{8.4}$$

Preservation gives $\dot{B}^a = v^a \approx 0$ and $\dot{\Pi}_a = -u_a \approx 0$, fixing the auxiliary multipliers. There is no new physical equation. For functions of the reduced variables alone, the Dirac bracket

$$\{A, B\}_{\mathrm{D}} = \{A, B\} - \{A, \Theta_I\}(\Delta^{-1})^{IJ}\{\Theta_J, B\} \tag{8.5}$$

equals the original reduced Poisson bracket. Spatial tensor transformations of the auxiliary labels can be restored off the constraint surface by adding terms proportional to $\Pi_a \mathcal{L}_{\boldsymbol{\xi}} B^a$; the reduced theory is unchanged.

Proposition 8.1 (An explicit canonical completion)**.** *The auxiliary system* (8.2) *together with the reduced gravitational constraints is a closed canonical realization of the physical constrained theory. Its symplectic reduction is exactly the phase space and Hamiltonian of Section 7. It adds no physical degrees of freedom.*

Proof. Equation (8.3) removes all twenty auxiliary phase-space coordinates. The reduced constraints commute with the auxiliary constraints in the displayed extension and obey Theorem 7.1. Thus

$$\frac{2(6+n+10) - 2\times 4 - 20}{2} = 2 + n. \tag{8.6}$$

The Dirac bracket calculation supplies the physical symplectic statement. □

The Π_a in this construction are auxiliary canonical coordinates. We have not asserted that they are the raw momenta conjugate to $b_{\mu\nu} - \chi g_{\mu\nu}$ in Eq. (3.2). Those raw momenta can include velocity mixing and additional primary chains. Proposition 3.1 and the pulled-back presymplectic form establish the physical equivalence needed here. A direct unreduced Dirac–Bergmann calculation would provide a further description, rather than a premise silently assumed in the proof.

Appendix D gives a homogeneous gravitational example in which an excluded-direction raw momentum is explicitly nonzero on $B = 0$. Thus replacing the raw primary constraints by $\Pi_a = 0$ would be unjustified. The proposition constructs an auxiliary canonical realization; it does not classify all raw primary, secondary, and higher constraints of Eq. (3.2).

This is why substitution alone is not the closure argument. The argument consists of variational equivalence, a physical symplectic identification, and a closed canonical evolution on that reduced space.

9 P2 and preservation of the real mass shell

9.1 Free probes on a dynamical background solution

Let (g, χ) be any smooth background solution obtained from the field theory. A free probe follows

$$S_{\mathrm{p}} = \int \mathrm{d}\tau \left[p_\mu \dot{x}^\mu - \frac{e}{2}\mathcal{Q}_{\mathrm{R}}(x, p) \right], \tag{9.1}$$

with x, p, e real and m fixed. The multiplier enforces $\mathcal{Q}_{\mathrm{R}} = 0$. Hamilton's equations are

$$\dot{x}^\mu = e g^{\mu\nu} p_\nu, \qquad \dot{p}_\mu = -\frac{e}{2}(\partial_\mu g^{\rho\sigma}) p_\rho p_\sigma. \tag{9.2}$$

Here the background may depend on the coordinate x^0; it has no independent explicit dependence on the worldline parameter τ. The equations preserve reality and the real mass shell. With an appropriate parameter choice they are the timelike or null geodesic equations of g.

Proposition 9.1 (P2 propagation)**.** *On their common zero set, away from the massless vertex, the real and imaginary parts of $\mathcal{Q}_{\mathbb{C}} = F^{-1}\mathcal{Q}_{\mathrm{R}}$ have a rank-one real differential. If $\mathcal{Q}_{\mathrm{R}} = 0$ initially, both parts remain zero along Eq. (9.1), for arbitrary smooth real χ.*

Proof. One has

$$\operatorname{Re}\mathcal{Q}_{\mathbb{C}} = \frac{\mathcal{Q}_{\mathrm{R}}}{1+\chi^2}, \qquad \operatorname{Im}\mathcal{Q}_{\mathbb{C}} = -\frac{\chi\mathcal{Q}_{\mathrm{R}}}{1+\chi^2}. \tag{9.3}$$

On $\mathcal{Q}_{\mathrm{R}} = 0$, the differentials of these functions are proportional to $\mathrm{d}\mathcal{Q}_{\mathrm{R}}$; the real coefficient is strictly positive. Thus the rank is one wherever $\mathrm{d}\mathcal{Q}_{\mathrm{R}} \neq 0$. The worldline Hamiltonian gives $\dot{\mathcal{Q}}_{\mathrm{R}} = 0$, and hence

$$\frac{\mathrm{d}\mathcal{Q}_{\mathbb{C}}}{\mathrm{d}\tau} = F^{-1}\dot{\mathcal{Q}}_{\mathrm{R}} + \frac{\mathrm{d}F^{-1}}{\mathrm{d}\tau}\mathcal{Q}_{\mathrm{R}} \approx 0. \tag{9.4}$$

No condition on $\dot{\chi}$ is generated. The two real parts also have a weakly vanishing mutual Poisson bracket. Writing $a = (1+\chi^2)^{-1}$, it is

$$\{a\mathcal{Q}_{\mathrm{R}}, -\chi a\mathcal{Q}_{\mathrm{R}}\} = -a^2\mathcal{Q}_{\mathrm{R}}\{\mathcal{Q}_{\mathrm{R}}, \chi\} \approx 0, \tag{9.5}$$

with the usual real cotangent bracket. This does not turn them into two independent first-class constraints. □

One regular first-class mass-shell constraint and its reparameterization freedom reduce the eight-dimensional particle phase space to six physical dimensions. Future orientation is preserved on the connected regular branch. The probe calculation does not include the distributional self-gravity of a point particle; the smooth field sources in Section 6 provide the backreacting matter system used in the Cauchy theorem.

The qualification "on the common zero set" is essential. If q denotes the value of $\mathcal{Q}_{\mathrm{R}}$ and $a = (1+\chi^2)^{-1}$, then

$$\det\frac{\partial(aq, -\chi aq)}{\partial(q,\chi)} = -\frac{q}{(1+\chi^2)^2}, \qquad \mathrm{d}(\operatorname{Re}\mathcal{Q}_{\mathbb{C}}) \wedge \mathrm{d}(\operatorname{Im}\mathcal{Q}_{\mathbb{C}}) = -a^2\mathcal{Q}_{\mathrm{R}}\,\mathrm{d}\mathcal{Q}_{\mathrm{R}} \wedge \mathrm{d}\chi. \tag{9.6}$$

The differential can therefore have rank two off the shell when $\mathrm{d}\mathcal{Q}_{\mathrm{R}}$ and $\mathrm{d}\chi$ are independent. On the shell it has rank one, and the strong identity $\operatorname{Im}\mathcal{Q}_{\mathbb{C}} + \chi\operatorname{Re}\mathcal{Q}_{\mathbb{C}} = 0$ ensures that the common zero set is exactly $\mathcal{Q}_{\mathrm{R}} = 0$. Off-shell rank two does not create a second physical mass-shell condition.

The real worldline action is also an essential premise. On the shell, $X_{\mathcal{Q}_{\mathbb{C}}} = F^{-1}X_{\mathcal{Q}_{\mathrm{R}}}$ in the complexified cotangent tangent space. This complex multiple need not be a real vector field. Equation (2.7) represents the constraint; it does not replace the real Hamiltonian $e\mathcal{Q}_{\mathrm{R}}/2$ as a generator with real parameter and real lapse.

9.2 Free field masses and the interacting theory

For a constant scalar background Φ_0 in a flat reference geometry, require $V(\Phi_0) = 0$ and $V_{,A}(\Phi_0) = 0$. Canonically normalizing the positive matrix $\mathsf{G}(\Phi_0)$ gives the free mass matrix

$$\mathsf{M}^2 = \mathsf{G}(\Phi_0)^{-1/2}\, V_{,AB}(\Phi_0)\, \mathsf{G}(\Phi_0)^{-1/2}. \tag{9.7}$$

If it is positive semidefinite, its real eigenmodes have $g^{\mu\nu}p_\mu p_\nu + m_a^2 = 0$, $m_a^2 \geq 0$. In the baseline $V = 0$, these modes are massless. This connects the field realization with P2 in its free regime.

The assumption about the Hessian is a vacuum stability condition, not a consequence of constraint closure. Nor does P2 require a general interacting wave packet on a varying background to satisfy an unchanged free dispersion relation. Such backgrounds can mix modes. The shared characteristic cone, which concerns the highest derivatives, is a separate statement proved next.

10 Characteristics and the local Cauchy problem

10.1 The common cone in the exact two-derivative model

Linearize Eq. (6.9) around a smooth, possibly inhomogeneous solution. In the equation with its target index lowered, the scalar principal symbol is

$$P_{AB}(k) = \mathsf{G}_{AB}(\Phi)\, g^{\mu\nu}k_\mu k_\nu. \tag{10.1}$$

Target connection terms contain first derivatives of the fields. Variation of the spacetime connection gives first derivatives of the metric perturbation, rather than second derivatives mixing the scalar and metric blocks. Hence

$$\det P(k) = \det \mathsf{G}\, [g^{\mu\nu}k_\mu k_\nu]^n. \tag{10.2}$$

For invertible G, the scalar characteristics are exactly the real metric light cone.

Fix generalized harmonic coordinates for the metric. The reduced Einstein equations have wave principal part $-\frac{1}{2}g^{\rho\sigma}\partial_\rho\partial_\sigma g_{\mu\nu}$. Their matter source in Eq. (6.8) has only first field derivatives. The coupled principal system is therefore a block wave system with the same g-cone. Its physical tensor and scalar modes share this cone once gauge and constraints have been treated.

This result is exact within Eq. (6.6), including the full response geometry. The preservation of the algebraic identity $\gamma = Fg$ and the characteristic statement are distinct parts of the result. An action with additional derivative interactions would need its own principal-symbol analysis.

10.2 Constraint propagation in harmonic gauge

Let C^μ denote the harmonic gauge vector, defined using a fixed smooth reference connection if needed. The reduced Ricci equation can be written

$$R_{\mu\nu} - \nabla_{(\mu}C_{\nu)} = M_{\mathrm{P}}^{-2}\left(\mathsf{G}_{AB}\partial_\mu\Phi^A\partial_\nu\Phi^B + g_{\mu\nu}V\right). \tag{10.3}$$

Combining the contracted Bianchi identity with Eq. (6.10), on solutions of the scalar equations, yields the homogeneous subsidiary equation

$$\Box_g C_\nu + R_{\nu\mu}C^\mu = 0. \tag{10.4}$$

Choose initial gauge data with $C|_\Sigma = 0$. The Einstein Hamiltonian and momentum constraints imply the vanishing normal derivative of C in the reduced system. Uniqueness for Eq. (10.4) then gives $C = 0$ throughout the local development. The reduced solution therefore solves the full Einstein equations.

The same preservation is expressed canonically by Theorem 7.1. The harmonic calculation also supplies the hyperbolic system needed for existence and continuous dependence; a bracket calculation alone would not establish those properties.

10.3 A local existence statement

Theorem 10.1 (Local classical development)**.** *Let Σ be a compact three-manifold without boundary. Prescribe initial data*

$$(h_{ij}, K_{ij}, \Phi^A, v^A), \qquad v^A = n^\mu \partial_\mu \Phi^A, \tag{10.5}$$

of Sobolev regularity $(H^s, H^{s-1}, H^s, H^{s-1})$, with an integer $s \geq 5$. Assume that h is positive definite, the constraints of Eqs. (7.4)–(7.5) hold, and the initial target image lies in a region with smooth uniformly positive G and smooth V. Then the system (6.6) has a local Lorentzian development, unique up to diffeomorphism, with continuous dependence on the initial data. In a fixed admissible harmonic gauge, (g, Φ) have the usual $C^0 H^s \cap C^1 H^{s-1}$ local regularity.

Reconstructing $\chi = \sinh(\phi/f)$ and $b = \chi g$ gives a solution of the multiplier theory on the regular branch. P1's geometric condition holds throughout this development, and the free probes of Section 9 preserve P2.

Proof. The harmonic metric equations and the scalar equations form a smooth quasilinear wave system with the common Lorentzian principal metric identified above. Local energy estimates for such systems give existence, uniqueness in fixed gauge, and continuous dependence at the stated, deliberately nonminimal regularity [5]. Positivity and nondegeneracy persist for a sufficiently short interval by continuity. The subsidiary equation restores the full Einstein equations. Geometric uniqueness follows by comparing harmonic gauges, using the standard Cauchy-development argument [6]. Finally, Proposition 3.1 reconstructs the multiplier solution; Eq. (9.4) proves the probe statement. □

The theorem applies the established local hyperbolic framework to the displayed CR action. It is not a new general existence theorem for Einstein–matter systems. The choice $s \geq 5$ avoids a claim about sharp regularity thresholds. On a compact initial slice, the finite field image and fixed $C_a > 1$ provide a positive lower bound for the target eigenvalues in a neighborhood of that image. The proof uses local coordinate patches and finite propagation; it does not assume that one global harmonic coordinate chart covers the compact manifold. Reconstruction preserves P1 within the constrained theory, rather than proving attraction of arbitrary complex metrics toward its allowed configurations.

Local development does not imply nonlinear stability, geodesic completeness, or global absence of singularities. Equation (5.2) itself has a singular boundary at $t = 0$. The critical sector $C = 1$ is excluded: the normalized coordinates in Eq. (6.2) are undefined there, while the unnormalized matrix on the right of Eq. (6.5) loses rank. This is not a uniform $C \to 1$ limit of the theorem. A separate constraint and Cauchy analysis in regular variables is required.

11 Closure does not uniquely select the response law

The preceding proofs used few details of Eq. (6.2). The Hamiltonian matter bracket required an invertible target metric. Positive definiteness supplied a healthy scalar kinetic sector and the chosen local evolution assumptions. This leaves room for different interacting theories.

11.1 A positive deformation

Consider a single normal sector and define, for a constant $\varepsilon > 0$,

$$\mathsf{K}_C^{(\varepsilon)}(\phi) = \mathsf{K}_C(\phi) + \varepsilon \frac{\phi^2}{f^2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \tag{11.1}$$

The added matrix is positive semidefinite for every real ϕ. Thus the deformed target is smooth and positive definite for fixed $C > 1$. It has the same reference value and first derivative,

$$\mathsf{K}_C^{(\varepsilon)}(0) = \mathsf{K}_C(0), \qquad (\mathsf{K}_C^{(\varepsilon)})'(0) = \mathsf{K}_C'(0), \tag{11.2}$$

but a different second derivative. Keep the geometric ϕ kinetic term, $f^2 = 3M_{\rm P}^2/2$, and $V = 0$ unchanged.

Both actions satisfy the same algebraic P1 condition, the same probe P2 construction, the same reduced constraint algebra, and the same characteristic-cone statement. Both have the same canonically normalized free massless spectrum at the reference vacuum. They nevertheless have different intrinsic target geometry.

11.2 A coordinate-invariant distinction

Use the curvature convention

$$R^A{}_{BCD} = \partial_C\Gamma^A_{DB} - \partial_D\Gamma^A_{CB} + \Gamma^A_{CE}\Gamma^E_{DB} - \Gamma^A_{DE}\Gamma^E_{CB}, \qquad R_{ABCD} = \mathsf{G}_{AE}R^E{}_{BCD}. \tag{11.3}$$

For $\mathrm{d}s^2 = \mathrm{d}\phi^2 + \mathsf{K}_{ij}(\phi)\mathrm{d}q^i\mathrm{d}q^j$, one obtains

$$R_{\phi i\phi j} = -\frac{1}{2}\mathsf{K}''_{ij} + \frac{1}{4}(\mathsf{K}'\mathsf{K}^{-1}\mathsf{K}')_{ij}. \tag{11.4}$$

At the reference vacuum of the undeformed model, define

$$k = R_{UVUV}, \qquad r_U = R_{\phi U\phi U}, \qquad r_V = R_{\phi V\phi V}. \tag{11.5}$$

Since the basis is orthonormal there, these components are sectional curvatures. Direct evaluation gives

$$k = \frac{1}{4f^2(C^2-1)}, \tag{11.6}$$

$$r_U = (1 + 2C - 2C^2)k, \qquad r_V = (1 - 2C - 2C^2)k. \tag{11.7}$$

Under Eq. (11.1),

$$\delta k = 0, \qquad \delta r_U = -\frac{\varepsilon}{f^2}, \qquad \delta r_V = 0, \qquad \delta\mathcal{R}_{\mathcal{T}}(0) = -\frac{2\varepsilon}{f^2}. \tag{11.8}$$

The last relation compares scalar curvature at the specified reference vacuum. It is independent of the choice of target coordinates. Thus the two theories, viewed with that identified vacuum and geometric scalar, are not related merely by a field-coordinate change preserving the reference data. Curvature enters the derivative interactions of a scalar theory and can be expressed through its on-shell amplitudes [7, 8].

Proposition 11.1 (Underdetermination by classical closure)**.** *Within the generally covariant scalar action class of Eq. (6.6), P1, P2 in their stated free regime, positive scalar kinetic terms, reduced classical constraint closure, and the common characteristic cone do not uniquely determine the normal response metric (6.2).*

Proof. Equation (11.1) is an explicit continuous family satisfying all those conditions. Equation (11.8) distinguishes its target geometries at the reference vacuum. □

This proposition does not test a further axiom imposing a particular holomorphic construction, internal symmetry, or variational selection rule; none has been assumed here. It identifies precisely where such an axiom would have work to do.

12 What the construction establishes

For the geometric core, the multiplier action makes the restriction of allowed variations explicit. Its solutions and physical symplectic structure agree with those of the restricted action. The exact complex-vacuum counterexample prevents a stronger, incorrect identification of parent equations.

With positive normal response sectors, the reduced theory has a closed classical gravitational constraint algebra, $2+n$ physical local modes, a shared characteristic cone, and a local Cauchy development. Free probe momenta retain their real mass shell and its complex representation, whose differential has rank one on the shell. These are properties of the full classical actions displayed here, rather than properties inferred from a small-field truncation.

Result	Precise content
Variational consistency	Exact equivalence of multiplier and restricted bulk equations on the regular branch.
Canonical closure	Explicit reduced ADM algebra and an auxiliary second-class realization with the same physical phase space.
P2 propagation	Preservation of the real and compatible complex shell for regular free probes.
Causal evolution	A common characteristic cone and local well-posed development for smooth positive normal sectors.
Selection boundary	A positive response deformation preserves closure while changing target curvature.

The remaining tasks have different mathematical forms. Selection of the response law requires an additional restriction that excludes deformations such as Eq. (11.1). A theory of the ambient complex spacetime must specify its independent fields away from M, its reality or analyticity conditions, and how it induces the slice dynamics. The critical $C = 1$ branch requires a new regular-stratum analysis. A quantum formulation must define physical states and a quantum implementation of the constraints, including any anomaly question. The classical brackets established here do not settle that operator problem.

The construction therefore supplies a definite classical starting point for a fundamental CR theory. Its next structural test is whether the complex geometry can select interactions more tightly than general covariance and a common light cone alone.

A Details of the conformal and cosmological calculations

For $\gamma = Fg$, the connection difference is

$$\Gamma^{\rho}_{\mu\nu}[\gamma] - \Gamma^{\rho}_{\mu\nu}[g] = \frac{1}{2F}\left(\delta^{\rho}_{\mu}\partial_{\nu}F + \delta^{\rho}_{\nu}\partial_{\mu}F - g_{\mu\nu}\nabla^{\rho}F\right). \tag{A.1}$$

Its contraction gives Eq. (5.4). Taking the γ-trace then gives

$$R[\gamma] = F^{-1}\left[R[g] - 3F^{-1}\Box F + \frac{3}{2}F^{-2}(\nabla F)^2\right], \tag{A.2}$$

which reproduces Eq. (4.1) after multiplication by the volume element.

For the exact solution (5.2)–(5.3), at $t = t_0$,

$$\dot{\chi} = \frac{2}{3t_0}, \qquad \ddot{\chi} = -\frac{2}{3t_0^2}, \qquad \Box\chi = 0, \qquad \nabla_i\nabla_j\chi = -\frac{2}{9t_0^2}\delta_{ij}. \tag{A.3}$$

Using $R_{tt}[g] = 2/(3t_0^2)$ in Eq. (5.4), the real contribution cancels $3\dot{\chi}^2/2$, leaving $2\mathrm{i}/(3t_0^2)$. The spatial component gives $2\mathrm{i}\delta_{ij}/(9t_0^2)$.

A second calculation writes the complex metric directly as

$$\gamma = -\mathfrak{n}^2 \mathrm{d}t^2 + \mathfrak{a}^2 \mathrm{d}\boldsymbol{x}^2, \qquad \mathfrak{n}^2 = F, \quad \mathfrak{a}^2 = Fa^2. \tag{A.4}$$

On the local branch at $F = 1$,

$$R_{tt}[\gamma] = -3\left(\frac{\ddot{\mathfrak{a}}}{\mathfrak{a}} - \frac{\dot{\mathfrak{a}}}{\mathfrak{a}}\frac{\dot{\mathfrak{n}}}{\mathfrak{n}}\right), \tag{A.5}$$

$$R_{ij}[\gamma] = \frac{\mathfrak{a}^2}{\mathfrak{n}^2}\left(\frac{\ddot{\mathfrak{a}}}{\mathfrak{a}} + 2\frac{\dot{\mathfrak{a}}^2}{\mathfrak{a}^2} - \frac{\dot{\mathfrak{a}}}{\mathfrak{a}}\frac{\dot{\mathfrak{n}}}{\mathfrak{n}}\right)\delta_{ij}. \tag{A.6}$$

These expressions independently reproduce Eq. (5.5).

B Initial constraints and the canonical normalization

Let $v^A = n^\mu \partial_\mu \Phi^A$. The initial energy density and scalar momentum contraction are

$$\rho = \frac{1}{2}\mathsf{G}_{AB}\left(v^A v^B + h^{ij}\partial_i\Phi^A\partial_j\Phi^B\right) + V, \qquad m_i = \mathsf{G}_{AB}v^A\partial_i\Phi^B. \tag{B.1}$$

In the extrinsic-curvature convention of Eq. (7.1), the canonical constraints become

$${}^{(3)}R + K^2 - K_{ij}K^{ij} = \frac{2\rho}{M_{\mathrm{P}}^2}, \qquad D_j(K^j{}_i - \delta^j_i K) = \frac{m_i}{M_{\mathrm{P}}^2}. \tag{B.2}$$

These are restrictions on initial data; they are not additional equations to impose independently at every time step after a constraint-preserving evolution has been established.

Tracing Eq. (7.2) gives $\pi = -M_{\mathrm{P}}^2\sqrt{h}\,K$, so

$$K^{ij} = \frac{2}{M_{\mathrm{P}}^2\sqrt{h}}\left(\pi^{ij} - \frac{1}{2}\pi h^{ij}\right). \tag{B.3}$$

The identity

$$\pi^{ij}\pi_{ij} - \frac{1}{2}\pi^2 = \frac{M_{\mathrm{P}}^4 h}{4}(K_{ij}K^{ij} - K^2) \tag{B.4}$$

checks the factor in Eq. (7.4). For matter,

$$P_A\dot{\Phi}^A - \mathcal{L}_\Phi = N\left[\frac{\mathsf{G}^{AB}P_A P_B}{2\sqrt{h}} + \frac{\sqrt{h}}{2}\mathsf{G}_{AB}h^{ij}\partial_i\Phi^A\partial_j\Phi^B + \sqrt{h}V\right] + N^i P_A\partial_i\Phi^A. \tag{B.5}$$

The scalar shift term has the same sign as the matter contribution to $\mathcal{D}_i$.

C Target curvature at the reference vacuum

For the warped block metric in Section 11, the nonzero connection coefficients are

$$\Gamma^\phi_{ij} = -\frac{1}{2}\mathsf{K}'_{ij}, \qquad \Gamma^i_{\phi j} = \Gamma^i_{j\phi} = \frac{1}{2}(\mathsf{K}^{-1}\mathsf{K}')^i{}_j. \tag{C.1}$$

Besides Eq. (11.4), the purely response components obey

$$R_{ijkl} = -\frac{1}{4}\left(\mathsf{K}'_{ik}\mathsf{K}'_{jl} - \mathsf{K}'_{il}\mathsf{K}'_{jk}\right). \tag{C.2}$$

At $\phi = 0$,

$$\mathsf{K}'_C(0) = -\frac{1}{f\sqrt{C^2-1}}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \qquad \mathsf{K}''_C(0) = \frac{1}{f^2}\begin{pmatrix}\dfrac{C}{C+1} & 0\\ 0 & \dfrac{C}{C-1}\end{pmatrix}. \tag{C.3}$$

These give Eqs. (11.6) and (11.7), and

$$\mathcal{R}_{\mathcal{T}}(0) = 2(k + r_U + r_V) = 2k(3 - 4C^2). \tag{C.4}$$

The deformation changes only $\mathsf{K}''_{UU}(0)$, by $2\varepsilon/f^2$, proving Eq. (11.8). This calculation is local in target space but does not use a small-field approximation to define either action.

D Why a raw excluded-direction momentum need not vanish

A homogeneous example makes the distinction in Section 8 explicit. Work per unit comoving spatial volume, with the appropriate gravitational boundary term, and write

$$\gamma = -\mathfrak{n}^2 \mathrm{d}t^2 + \mathfrak{a}^2 \mathrm{d}\boldsymbol{x}^2, \qquad \mathfrak{n}^2 = N^2 + \mathrm{i}d, \qquad \mathfrak{a}^2 = q + \mathrm{i}b, \quad q > 0,\ N > 0. \tag{D.1}$$

Here b is the common spatial imaginary component and $-d$ the temporal one. The homogeneous complex Einstein–Hilbert kinetic Lagrangian is

$$L_0 = -\frac{3M_{\mathrm{P}}^2}{4}\,\mathrm{Re}\,\frac{(\dot{q} + \mathrm{i}\dot{b})^2}{\sqrt{N^2 + \mathrm{i}d}\sqrt{q + \mathrm{i}b}}. \tag{D.2}$$

The square roots use the sheet continuous from positive real N and $\sqrt{q}$. P1 requires $b = \chi q$ and $d = \chi N^2$. Substituting these relations, including $\dot{b} = \chi\dot{q} + q\dot{\chi}$, yields

$$L_{\mathrm{red}} = -\frac{3M_{\mathrm{P}}^2\dot{q}^2}{4N\sqrt{q}} + \frac{3M_{\mathrm{P}}^2 q^{3/2}\dot{\chi}^2}{4N(1+\chi^2)}. \tag{D.3}$$

With $q = a^2$ and $\phi = f\,\mathrm{arsinh}\,\chi$, this is precisely the flat homogeneous Einstein–scalar action. The lapse is retained during variation, so its Friedmann constraint is not lost.

At an event with $\chi = 0$, allow $\dot{\chi} \neq 0$. Differentiating the unreduced Lagrangian with respect to $\dot{b}$, then imposing the constraint, gives

$$p_b\big|_{\chi=0,B=0} = \frac{3M_{\mathrm{P}}^2}{2N\sqrt{q}}\dot{b} = \frac{3M_{\mathrm{P}}^2\sqrt{q}}{2N}\dot{\chi}, \qquad B = b - \chi q. \tag{D.4}$$

This is generally nonzero. Under the adapted coordinate change to B, the raw momentum conjugate to B equals p_b. By contrast, the momentum induced along the allowed χ direction is

$$p_\chi = q p_b = \frac{3M_{\mathrm{P}}^2 q^{3/2}}{2N}\dot{\chi} \quad \text{at } \chi = 0, \tag{D.5}$$

in agreement with Eq. (D.3). Thus the pullback of the canonical one-form works without setting the excluded-direction raw momentum to zero. This example is a check of the distinction, not a replacement for the full unreduced Dirac–Bergmann analysis.

E Reproducibility and limits of the checks

With Python 3.12.14 and SymPy 1.14.0, `check_closure.py` passed all **47 named checks** and `audit_closure.py` passed all **28 named checks**. The package includes both scripts, text and JSON results, an audit report, and rebuilding instructions. These are assertion counts, not independent predictions or proofs.

The original checks cover light-cone rigidity, conformal reduction, target curvature and its deformation, ADM normalization, the matter bracket, auxiliary constraints, mass-shell identities, and the cosmological counterexample.

The audit adds mechanical multiplier and homogeneous gravity checks, nonorthogonal ADM contractions, the gravitational lapse-derivative coefficient, and a functional matter bracket with a nondiagonal target in all three spatial directions. It also checks an off-shell Noether identity, shell versus off-shell rank, real versus complex Hamiltonian flow, and all complex FLRW Ricci components and the reaction multiplier at the reference event. Mechanical examples and pointwise checks do not replace general proofs.

These calculations do not complete the raw Dirac chains, prove PDE existence, establish quantum closure, or test observations. The variational and canonical proofs are in the text; local existence applies the stated classical hyperbolic framework.

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